

RDC8102 driver

Overview

The RDC8102 protocol for Adroit communicates to a Reporter 8102, which is designed to receive transmitter codes in a control room or operate as a stand alone intelligent repeater. Product obtainable from Radio Data Communications (Tel 011 616 7685)

The protocol is an exception based protocol driver in that data is received asynchronously when a value changes on an outstation.

Wiring

Wiring between the RS232 port and the modem is standard RS232. Please consult your REPORTER 8102 documentation for additional particulars on setting up the hardware.

Driver Installation and Setup

The driver is installed by picking the RDC8102 driver from the drivers directory. The RDC8102.DLL and the RDC8102.DOC is then copied into the agent directory.

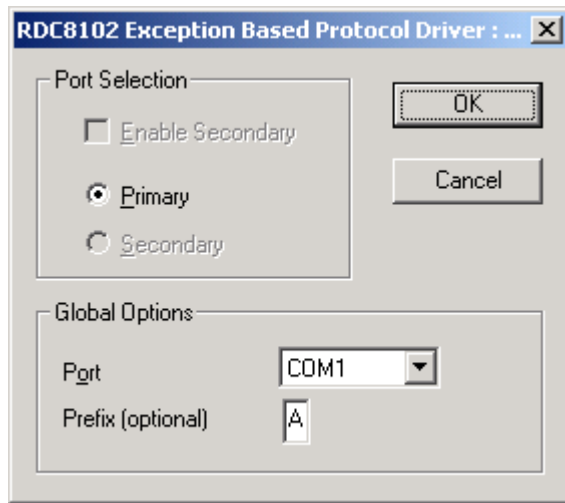
After installation of the driver, the user may then create and configure devices for the RDC8102 driver.

Device Configuration

Devices

At a minimum, you must create at least one device for every port being used. However, you may choose to create a device for a group of transmitters.

Device configuration dialog



Port	Select the port through which you are going to communicate. The baud rate is fixed at 9600,8,1,N.
Prefix	This field is only used when the protocol is configured to communicate in alpha prefix mode. In all other modes this field is ignored.

Scanning

Due to the system operation, all scanned tags must be output enabled, pulsed, and cold started to zero. The reasoning follows: The scanning sub-system keeps a record of the last received value for optimization, and thus will not re-send an alarm “on” until it goes “off” again. By output enabling the tag, when the tag goes to “zero” the scanning system updates it’s last known value with the zero (“off”), thus when the alarm on (“one”) condition arrives again, it gets put into the tag as expected. By the same reasoning, cold-start will initialize the tags to zero (“off”), so that all current alarms are clear. In addition, stopping and starting a device agent causes the driver to zero all scanned values, and the tags.

The driver keeps a last known value for the alarms, which is cleared on device start, this last known value gets “OR”ed with an incoming alarm, thus giving the SUM of alarms received to the tag. Effectively this “copy” of the value allows the driver to keep track of the tags’ value. This function naturally only works correctly if the tag was output enabled, so that when bits in the tag clear (go off) the driver will also clear them at the same instant in time.

Outstation Data

The driver processes the following data received:

0	-	Alarm
1	-	Lock up
2	-	Open up
3	-	Emergency
4	-	Telemetry 1
5	-	Telemetry 2
6	-	Telemetry 3
7	-	Telemetry 4
8	-	Mains Fail

9	-	Mains Restore
10	-	Engineer Test
11	-	Customer Duress
12	-	Medical Alert
13	-	Fire Alarm
14	-	Telemetry 5
15	-	Telemetry 8

Data received in any mode will be processed according to the above list. In Chubb mode only bits(0-7) will be activated.

Digital inputs

Digital tags (Boolean slots) that are scanned from Outstations can be scanned into Digital tags

The format of the address string is as follows: CN: TTTT: B

Where: CN – stands for code number.

TTTT – Code number of transmitter

B – Bit number (0 - 15)

Analogue / Multistate inputs

If scanned into a Multistate agent than the 16 slots represent the digital status of that transmitter.

The format of the address string is as follows: CN: TTTT

Where: CN – stands for code number.

TTTT – Code number of transmitter

When scanned into an integer, the value of the integer will represent the status of the outstation by a number ranging from 1 to 65535 with the value 0 representing no alarms.

Message protocol

Datascope messages are displayed in hex or ascii. All messages are displayed whether that transmitter is configured or not. Messages of devices not configured are displayed in the first devices protocol monitor.

<u>Reporter Standard</u>	Code Number	Telemetry	Repeater	CR
Byte Number	1- 4	5	6	7
Example	1220	1	1	^13

<u>3 Digit Telemetry</u>	Code Number	Telemetry	Repeater	CR
Byte Number	1- 4	5-7	8	9
Example	1220	012	1	^13

<u>Chubb Format</u>	Code Number	Space	Telemetry	CR
Byte Number	1- 4	5	6	7
Example	1220		1	^13

Protocol with alpha prefix – Any of the above three protocols with an alpha prefix. Thus this protocol will have 1 byte more than the above 3 protocols. For the data to be valid the prefix defined must match the prefix defined in the device.

Troubleshooting

Revisions

Date	Changes
91/02/2002	Dialog updated after neatening up.
24/02/2003	Described better output enable and cold-starting of the tags.